

Mistaking Machine Style for Sarcasm: Diagnosing Class-Conditional Negative Transfer in Joint Bilingual Sarcasm and AI-Authorship Detection

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Abstract

A deployed system may have to judge, for the same text, both whether it is sarcastic and whether a human wrote it, yet the two detection tasks are almost always studied apart. We train them jointly in a bilingual Chinese–English setting and build DualTest (4,664 instances), a diagnostic test set stratified by source and sarcasm label; labels on the AI side reflect the requested generation style rather than independent human annotation. Joint training turns out to have a sharply localized cost. Relative to a single-head sarcasm model with the same supervision, adding an authorship head raises the false-positive rate on AI-generated non-sarcastic text from 19.3% to 86.3%. The failure replicates with a second generator (91.3%), and two intermediate architectural modules reduce it substantially. Probing shows that source information is decodable and associated with sarcasm decisions; a control that leaves source information intact, however, reproduces the same effect, so we characterize the failure as a source-correlated, style-entangled shortcut, with the source concept’s specific causal role remaining a hypothesis. Detector scores also diverge from blinded LLM-judge evaluations—one more reason not to use authorship detectors directly as evaluation or reward signals for sarcasm generation.

1 Introduction

A sarcastic sentence usually means something other than what it literally says, and often the opposite. With few stable lexical cues to anchor on, sarcasm detection has long depended on context, intent, and background knowledge; annotation agreement is relatively low, and multilingual settings are harder still. LLMs bring a further complication. Sarcastic social-media text may now be written by a person or generated by a model under instruction, so a deployed system can face “is this sarcastic?” and “was this written by a human?” on the same

input at the same time. The generation side couples the two questions as well: detector scores are already used in practice as evaluation and even reward signals for sarcasm generation (e.g., ViSP; Wang et al., 2026; §2.4), and judging whether generated sarcasm reads as human-written involves both dimensions at once. Existing work nonetheless treats sentiment/rhetorical analysis and content provenance/safety detection separately, and to our knowledge no prior work unifies sarcasm detection and AI-authorship detection as a joint multi-task learning problem (§2.2).

What can go wrong when these two judgments share a representation? When a single textual backbone supports both source identification and semantic judgment, the representation shaped by one task may alter how the other behaves on particular subgroups, whatever the architecture. We work with a dual-head model because it keeps this risk observable and localizable at minimal structural cost. Jointly trained with the authorship task, the sarcasm head systematically misclassifies AI-generated non-sarcastic text as sarcastic—a form of class-conditional negative transfer. The error is induced by joint training; it persists when the AI test texts are regenerated by a second generator, and intermediate modules mitigate it. At the mechanism level, probing shows that source information is highly decodable from the shared representation, correlated with sarcasm decisions, and aligned with the decision direction in whitened geometry. Concept-erasure experiments, however, locate the effect in a dominant-variance subspace entangled with stylistic factors (§5.3.2, Appendix E). The claim that the sarcasm head exploits a source-correlated shortcut is thus a mechanism hypothesis with preliminary support; confirming it will require causal mediation and nonlinear tools, which we leave to dedicated study. An orthogonal control makes concrete the gap between a decodable, decision-correlated signal and an estab-

lished mechanism, in line with methodological critiques of probing (Hewitt and Liang, 2019; Elazar et al., 2021).

Contributions. We formulate joint Chinese–English sarcasm and AI-authorship detection and introduce DualTest, which supports subgroup analysis across four source $\times$ sarcasm quadrants; this is the first study, to our knowledge, to examine the two tasks jointly at this granularity. Controlled single-head, dual-head, architectural, and cross-generator comparisons then reveal a class-conditional negative-transfer failure concentrated on AI-generated non-sarcastic text. Finally, representation analyses characterize the failure as source-correlated and style-entangled, and a detector–judge comparison shows that detector scores may be unreliable as standalone generation-evaluation signals.

2 Related Work

2.1 Sarcasm Detection and Multilingual Resources

Feature engineering (Riloff et al., 2013; Ghosh and Veale, 2016), neural sequence models (Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each pushed sarcasm detection forward, but the task stays hard in multilingual, intent-oriented, class-imbalanced settings: the best English systems on iSarcasmEval (Abu Farha et al., 2022) reach sarcasm F1 of roughly 0.56–0.61, zero-shot LLMs on MultiPICO (Casola et al., 2024) 0.15–0.57, and SarcNet (Yue et al., 2024) 0.73. Yang and Ikeda (2024), building on their earlier multilingual mBERT/BiLSTM study (Yang and Ikeda, 2023), introduced mBERT-RCNN (mBERT with the recurrent convolutional text-classification architecture of Lai et al., 2015) and ChatGPT data augmentation, finding that the RCNN’s local feature extraction improved robustness on mixed datasets of typologically distinct languages, an architectural line this work follows (§4.1). Because those results come from different test sets, §5.1 provides directly comparable mBERT dual-head baselines on our test set. Our setting is parallel Chinese–English; what existing resources lack, and what DualTest supplies, is evaluation data labeled jointly for sarcasm and AI authorship.

2.2 AI-Generated Text Detection and Authorship Verification

AI-text detection uses zero-shot/perplexity-based detectors (Mitchell et al., 2023) and discriminative models trained on paired human–machine corpora (HC3; Guo et al., 2023); generalization is the main difficulty; M4 (Wang et al., 2024) targets black-box detection across generators, domains, and languages, and RAID (Dugan et al., 2024) stress-tests detector robustness. Conceptually, Bevendorff et al. (2025) frame the problem as authorship verification rather than quality assessment; we treat the authorship judgment as an auxiliary task within a shared encoder. Existing multi-task sarcasm research mainly combines sarcasm with sentiment, stance, or emotion. We found no prior study that jointly trains sarcasm and AI-authorship prediction under a shared encoder;¹ that gap is the starting point of this study.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer direction depends on task relatedness and data geometry (Zhang et al., 2023; Wu et al., 2020), and conflicting gradients can cause negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide the closest NLP precedent, showing through single-/multi-task contrasts that models may infer task identity from the input distribution and destabilize on out-of-distribution inputs (see also He and Choi, 2021; Li et al., 2023, on internal mechanisms of multi-task models). Shortcut learning offers a complementary reading: models may exploit features that are predictive in the training distribution yet fail on specific subgroups (Geirhos et al., 2020; Ye et al., 2026); in multi-task settings this shows up as increased absorption of spurious correlations (Hu et al., 2022b; Chai et al., 2025) and as subgroup-stratified failures (Compton et al., 2023). Mitigation under a shared encoder takes two broad forms: some methods isolate task-specific parameters (AdapterFusion, Pfeiffer et al., 2021; MAD-X, Pfeiffer et al., 2020; HyperFormer++, Karimi Mahabadi et al., 2021; PALs, Stickland and Murray,

¹Searched in May 2026 across Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, using keyword combinations including “sarcasm detection,” “AI-generated text detection,” “authorship detection,” “multi-task learning,” and “joint detection.”

2019), while others filter interference inside the shared representation through gating, attention, or intermediate modules (Wu et al., 2019; CA-MTL, Pilault et al., 2021); our intermediate-layer analysis belongs with the second group. In this paper, conditional stratification exposes a subgroup-specific failure induced by joint training, and controlled comparisons of three intermediate structures test the mitigation hypothesis (§5.3).

2.4 Misalignment in Detector-Based Evaluation and Rewards

Automated scorers are increasingly used as evaluation or reward signals for generation (Zheng et al., 2023); optimization can then end up targeting a model-specific decision boundary rather than perceived quality. Sarcasm generation already runs this risk. ViSP (Wang et al., 2026) uses the DIP sarcasm detector (Wen et al., 2023) as a PPO reward (Schulman et al., 2017) and acknowledges dependence on evaluator quality. Section 6 tests whether our detector scores agree with independent blinded LLM judgments.

3 Dataset Construction

We begin with five heterogeneous sources (iSarcasm: Oprea and Magdy, 2020; Abu Farha et al., 2022; CSC: Jang and Frassinelli, 2024; Jang et al., 2023; ToSarcasm: Liang et al., 2022; News Headlines v2: Misra and Arora, 2023; Misra and Grover, 2021; Open Chinese Internet Sarcasm Corpus: Zhu, 2022; 44,862 instances after standardized merging, approximately 87% English) and obtain three training sets and one diagnostic test set through a three-stage pipeline. Source-level details, annotation methods, and quality checks are provided in Appendix A.

Sarcasm classification training set, SarcTrain (10,000, all human). The dataset is fully balanced across four language $\times$ label buckets, with 2,500 instances per bucket through downsampling, and is pre-split into five folds. All texts in this set are human-authored.

Authorship classification training set, AuthTrain (19,995). The human side is sampled directly from SarcTrain with language balance. On the AI side, one same-topic version is generated for each human instance using deepseek-r1:8b (DeepSeek-AI, 2025, via Ollama): paired AI versions were successfully obtained for 9,996 of 9,999

human instances, while three generation failures were retained only on the human side. Generation instructions are selected from four templates (Chinese/English $\times$ sarcastic/normal) according to the seed instance’s sarcasm label, so that the AI texts likewise cover both sarcastic and non-sarcastic styles and the authorship label is independent of sarcastic style. Generation parameters, full templates, and post-processing steps are given in Appendix A.

Generation training set, GenTrain (2,428; used in §6). This set is drawn from two sources containing topic or contextual information (iSarcasm and ToSarcasm; see Appendix A for their uses). Only positive sarcasm instances are retained; approximately 40% are sampled, and the minority language is oversampled to 1,214 instances in each of Chinese and English. By prompt type, the set contains 374 rewriting, 839 response, and 1,215 comment instances, and serves as the instruction-tuning data for SFT.

Diagnostic test set, DualTest (4,664). DualTest combines the official iSarcasm test split (1,400 English instances) and the ToSarcasm test set (972 Chinese instances) with AI-generated counterparts produced by deepseek-r1:8b. Human instances retain their original sarcasm annotations. For AI instances, `is_human` is false, while the sarcasm label denotes the style requested by the corresponding “sarcastic style” or “normal style” generation instruction (full templates in Appendix A). For this reason we treat the AI-side labels as instruction-conditioned diagnostic labels rather than independently annotated judgments of perceived sarcasm. The four source $\times$ sarcasm groups contain 823 human-sarcastic, 1,549 human-non-sarcastic, 779 AI-sarcastic, and 1,513 AI-non-sarcastic instances (see Appendix A for the three-dimensional distribution). The dataset preserves the natural composition of the original test splits.

4 Dual-Head Detection Model

4.1 Tasks and Architecture

Given a Chinese or English input text x , the model simultaneously produces two binary judgments: whether x conveys sarcastic intent and whether x was written by a human. The two tasks share a single DeBERTa-v3-large encoder (He et al., 2023, this encoder was pretrained on English corpora, and §5.1 directly compares it with multilingual-

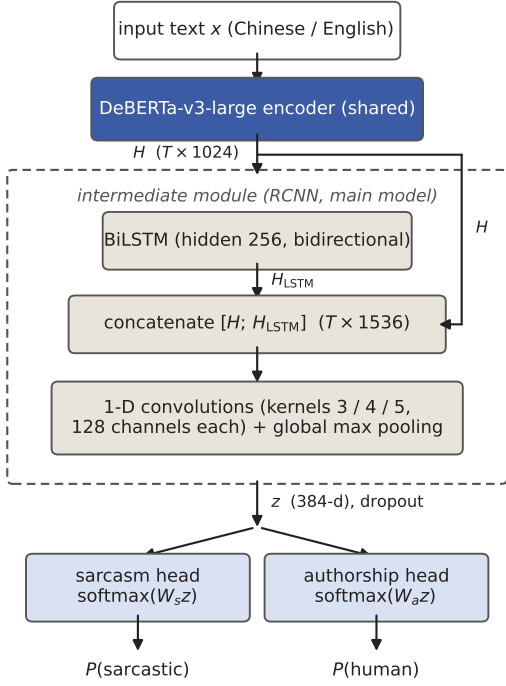


Figure 1: Main dual-head model: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The controlled comparisons in §5 vary only the dashed module: it is removed in the plain dual-head and replaced by task-branch interaction projection in the variant (Appendix B).

pretrained mBERT / XLM-R on the same bilingual test set). Encoder outputs are aggregated by an intermediate module and then passed to two independent classification heads; Figure 1 shows the main model.

The main model uses BiLSTM + CNN (RCNN) as its intermediate module: the encoder output sequence passes through a single-layer bidirectional BiLSTM (hidden size 256), is concatenated with the original sequence, and is aggregated into a 384-dimensional vector z by three groups of one-dimensional convolutions (kernel widths 3/4/5, 128 channels each) with global max pooling; after dropout, z is passed to the two heads. This intermediate module follows the mBERT-RCNN architectural idea of Yang and Ikeda (2024). The controlled comparisons in §5 include two additional variants. The plain dual-head model removes the intermediate module and passes the [CLS] representation directly to both heads. The task-branch interaction projection model replaces RCNN with two 256-dimensional task-branch projections of

the [CLS] representation, h_s and h_a , each augmented by a residual attention read of the opposite branch: $h'_s = h_s + \text{Attn}(Q=h_s, K=h_a, V=h_a)$, and symmetrically for h'_a . Because each branch consists of a single vector, the attention weight is identically 1 and the module reduces to a residual addition of a linear projection from the opposite branch; we name the variant by this actual computation rather than as a form of cross-attention (hereafter, the interaction-projection dual-head model). The encoder and classification heads are identical across all three models; only the intermediate structure differs, which is what makes the comparison of the intermediate layer’s role a controlled one. Remaining dimensions and dropout settings are in Appendix B.

4.2 Training Protocol

Training uses a mixed loader that alternates batches by task: each batch contains samples from only one task, and authorship and sarcasm batches alternate in a 2:1 ratio, matching the data-size ratio between AuthTrain and SarcTrain. Let the shared parameters be θ_{sh} (the encoder and intermediate module), let $z(x)$ be their aggregated representation for input x , and let W_s and W_a denote the two classification heads. At step t , the loss is computed only for the head corresponding to the task of the current batch B_t :

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(\sigma(W_s z(x)), y), & B_t \subset \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(\sigma(W_a z(x)), y), & B_t \subset \text{AuthTrain}, \end{cases} \quad (1)$$

where σ denotes softmax and CE_w is class-weighted cross-entropy that compensates for the scarcity of sarcastic positives (weights are provided in Appendix B). The gradient flow follows from the equation: on sarcasm batches, $\partial \mathcal{L} / \partial W_a = 0$ (symmetrically, $\partial \mathcal{L} / \partial W_s = 0$ on authorship batches), whereas θ_{sh} receives gradients from both types of batches.

Optimization uses separate learning rates for parameter groups: the pretrained encoder is updated at a much lower rate than the intermediate module and classification heads to preserve pretrained representations. Model selection accounts for both tasks: the early-stopping signal at epoch e is the equally weighted sum of each task’s validation loss normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and authorship tasks on their respective validation sets. Normalization prevents early stopping from being dominated by the larger authorship task.

Training and evaluation use five fold-based runs with a fixed external test set. SarcTrain is pre-split into five folds: run k trains the sarcasm task on four folds, with the held-out fold as the sarcasm validation set providing $\mathcal{L}_s^{(e)}$; AuthTrain uses a fixed 90% training / 10% validation split shared by all runs, its validation portion providing $\mathcal{L}_a^{(e)}$, so between-run variation is concentrated in changes to the sarcasm training data. Each run is trained from a fresh initialization of the same pretrained base; all architectures and ablations share the fold assignments and seed scheme, and cross-model comparisons are paired by split. The five models are then evaluated on both tasks on the external test set DualTest, which is involved in no split. The main tables report the mean $\pm$ standard deviation across the five runs (hereafter, five-run mean $\pm$ standard deviation); this standard deviation reflects between-run variation from training splits and initialization, while test-set sampling uncertainty is quantified separately through confidence intervals for the final model (Table 3 and Appendix C.7).

A separate final model is retrained on all of SarcTrain and AuthTrain using the same hyperparameters and a fixed epoch count determined from the five validation runs. It is used only for subgroup inference and §6 rescoring and is not selected on DualTest. It does not enter the five-run aggregate comparisons in Table 2 and Table 1; its quadrant estimates are reported separately in Table 3. Class weights, sampling ratio, grouped learning rates, and full hyperparameters are provided in Appendix B.

5 Experiments and Diagnosis

The experiments compare encoder backbones, joint versus single-task training, and intermediate-layer architectural variants on overall F1, then use DualTest’s dual labels to partition the same evaluation into four “source $\times$ sarcasm” quadrants, where a class-conditional failure emerges that overall F1 does not show. Aggregate model comparisons report the mean $\pm$ standard deviation over five fold-

based runs. The quadrant analysis uses the separately trained final model to obtain one prediction per instance for paired subgroup tests; Wilson intervals quantify test-set uncertainty. The final model is not selected on DualTest.

5.1 Backbone Comparison

We compare dual-head models across four encoder backbones (mBERT, XLM-RoBERTa-base, and DeBERTa-v3 base and large) and, on the large encoder, the two intermediate-module variants of §4.1. The encoders span multilingual and English-centric pretraining: mBERT is BERT-base pretrained on multilingual Wikipedia (Devlin et al., 2019), XLM-RoBERTa-base is pretrained on CommonCrawl text covering about one hundred languages (Conneau et al., 2020), and DeBERTa-v3 combines disentangled attention with ELECTRA-style pretraining (Clark et al., 2020) on primarily English corpora. “Sarcasm F1” is the F1 of the sarcastic class (positive class), following the official iSarcasmEval convention, and “Authorship F1” is the F1 of the human class (positive class); both are computed over the full DualTest set, aggregating Chinese and English and both source sides, with AI-side sarcasm labels constructed from the generation instructions (§3).

Table 1 reports the comparison (five-run mean $\pm$ standard deviation); the main model reaches sarcasm F1 0.6309 ± 0.020 and authorship F1 0.9301 ± 0.005 . Under the parameter scales and hyperparameter settings used here, DeBERTa-family encoders clearly outperform mBERT and XLM-R; the comparison mixes differences in parameter scale and pretraining corpora, and we use it only to pick the main model for the diagnosis. The two intermediate-layer variants differ little from the plain dual-head model in their means (with the main RCNN model performing better), but reduce between-run variance by approximately 34–47%. Chinese sarcasm F1 is substantially higher than English sarcasm F1 (0.819 vs. 0.429 for the main model), primarily because of differences in sarcasm prevalence between the two language subsets (63.8% vs. 14.2%) and differences in source-benchmark difficulty (Appendix C.7). Sarcasm F1 is substantially lower than authorship F1 for all models, and its absolute level of about 0.63 lies within the range of the multilingual, intent-oriented benchmarks cited in §2.1. Macro-averaged, human-subset, and per-language results for the main model are in Appendix C.

Model	Sarc. F1	Auth. F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLm-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.004
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN (main model)	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Dual-head detection performance across backbones and architectural variants (DualTest; five-run mean $\pm$ standard deviation; bold = main model).

Arch.	Task	Dual	Single	Δ
Plain	Sarc.	0.6126	0.6734	-6.08
Plain	Auth.	0.9221	0.9054	+1.67
RCNN	Sarc.	0.6309	0.6751	-4.43
RCNN	Auth.	0.9301	0.9253	+0.49

Table 2: Dual-head vs. single-head models (mean over five fold-based runs; Δ = dual - single, in percentage points).

5.2 Dual-Head versus Single-Head Ablation

Each comparison in Table 2 is made within the same architecture: a dual-head model and its corresponding single-head model share the encoder, intermediate structure, and training data, differing only in whether training is joint with the other task. Comparisons are paired run by run using the shared fold assignments (§4.2). Joint training lowers sarcasm F1 and slightly improves authorship F1. Paired run-level differences provide a more detailed picture than the means: all ten paired differences on the sarcasm side are consistently negative across the two architectures (-0.8 to -8.6 pp), whereas differences on the authorship side are centered around zero and are predominantly positive; the asymmetry does not depend on the architecture. RCNN also compresses the magnitude at both ends toward zero, and we read this as the intermediate layer screening out part of the inter-task interference. In the quadrant analysis below, this compression of the aggregate effect and the mitigation of the subset failure appear on the same two intermediate modules.

5.3 Four-Quadrant Diagnosis: Class-Conditional Negative Transfer and Intermediate-Layer Buffering

Aggregate metrics obscure the structure of errors. The overall sarcasm F1 of the plain dual-

head model alone would read as unremarkable mid-range performance; in fact its errors concentrate on the AI-text subset. To localize this class-conditional failure, we partition DualTest by the ground-truth `is_human` $\times$ `is_sarcastic` labels into four quadrant subsets and examine the detector’s sarcasm predictions separately within each. Because each quadrant contains only one ground-truth class, F1 degenerates on such subsets (precision is always 1 on positive-class subsets, while negative-class subsets contain no positive instances). We therefore report recall for positive-class subsets and the false-positive rate for negative-class subsets.

The single-task control and cross-generator replication establish the behavioral conclusion—class-conditional negative transfer exists and is induced by joint training—while representation-level tests supply correlational evidence for the shortcut hypothesis and locate the signal within the dominant-variance structure of the representation.

5.3.1 Three Architectures and Single-Head Controls

We compare the sarcasm predictions of five detectors on the same DualTest set. The three dual-head variants (plain / interaction-projection / RCNN) characterize how the severity of the failure changes with intermediate structure. The two single-head models share the same architecture and sarcasm supervision data (SarcTrain) as their corresponding dual-head models. They differ only in whether the auxiliary authorship task is jointly trained (that task’s training objective and the AuthTrain data are added as a whole). The single-head model supplies the counterfactual: the same sarcasm classifier without joint training. Were the failure rooted in the sarcasm training data (SarcTrain contains only human text), the single-head model would fail in the same way; a failure induced by joint training, by contrast, should leave it untouched. Table 3 reports sarcasm predictions. The failure is concentrated in the sarcasm head; the authorship head remains stable under the same partition (reported below and in Appendix C).

The plain dual-head model fails severely in the AI $\times$ non-sarcastic quadrant: it misclassifies 86.3% of the instances as sarcastic, indicating that on this subset the model likely relies on cues associated with AI source rather than stable sarcastic semantics. Both structurally distinct intermediate modules substantially mitigate the fail-

Model	H sarc. rec.	H non-sarc. FP	AI sarc. rec.	AI non-sarc. FP
Plain dual-head	67.1%	32.9%	98.2%	86.3% [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9% [38.5, 43.4]
RCNN dual-head (main)	80.8%	40.5%	89.3%	34.1% [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3% [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3% [17.4, 21.4]

Table 3: Comparison of five sarcasm classifiers on the four ground-truth quadrant subsets (recall for sarcastic subsets; false-positive rate, FP, for non-sarcastic subsets). Brackets in the final column give Wilson 95% confidence intervals over test instances. Intervals for all quadrants, paired tests on the same texts, and per-quadrant score distributions (showing that the failure is an overall rightward shift of the score distribution rather than a threshold artifact) are provided in [Appendix C.7](#).

ure, reducing it by more than 45 pp from 86.3%, with mutually non-overlapping confidence intervals across the rates (Table 3). The approximately 7 pp gap between RCNN and interaction projection is smaller but remains significant (Appendix C.7). The single-head models settle the counterfactual: despite also being trained only on human text, they have a false-positive rate of just 19.3%. The difference from the plain dual-head model on the same texts is significant by a paired McNemar test (discordant pairs 1,021 / 7, $p < 0.001$; exact p values and the remaining pairs are reported in Appendix C.7). This rules out missing AI text in the sarcasm training data as an explanation: the failure is induced by joint training. The RCNN and plain single-head models also behave almost identically (both approximately 19.3%): RCNN’s mitigating effect emerges only in the presence of joint training, not from inherent architectural robustness to AI text.

The authorship head provides a control for the failure’s directionality. On the same AI $\times$ non-sarcastic subset, its false-positive rate is only 6.1% (plain) / 6.6% (RCNN), and it remains stable across all four quadrants; the single-head authorship counterfactual performs at the same level as the dual-head models (complete values in Appendix C). On the same set of AI texts, the problem is concentrated in the sarcasm head and the gap is on the order of tens of percentage points. The failure is jointly conditioned on source and class and is most extreme on the AI $\times$ non-sarcastic subset, hence the term “class-conditional negative transfer.” This subset-level failure and the asymmetric aggregate transfer in §5.2 are two views of the same negative transfer.

5.3.2 Mechanism Explanation and Levels of Evidence

Because authorship gradients update the shared representation whereas SarcTrain contains only human texts, we hypothesize that the sarcasm head exploits source-correlated style cues that are not corrected by sarcasm supervision on AI text. Under this interpretation, an intermediate module may partially buffer changes in the shared representation. This account is a hypothesis rather than an established causal mechanism.

The representation-level evidence is mixed: source is highly decodable and correlated with sarcasm decisions, and LEACE (Belrose et al., 2023) reduces both source decodability and the AI $\times$ non-sarcastic false-positive rate. However, removing a high-variance direction orthogonal to source produces a comparable reduction while leaving source decodability intact, whereas ten random-direction placebos leave the rate unchanged. The effective interventions thus pick out a dominant-variance, style-entangled subspace but do not establish source identity as the causal feature, which is why we stay with the term source-correlated shortcut. Full probing, erasure, and control results are reported in Appendix E.

5.3.3 Cross-Generator Replication

We regenerate the AI side of DualTest with qwen3.5:9b (Qwen Team, 2026) while keeping the human side and trained detectors unchanged; Figure 2 plots the five detectors’ rates on both generators with Wilson 95% confidence intervals, and Appendix C.7 tabulates them with paired tests. The plain dual-head false-positive rate on AI-generated non-sarcastic text increases from 86.3% to 91.3%, whereas the corresponding single-head rate decreases from 19.3% to 7.9%. Mitigation is also preserved for RCNN (34.1% $\rightarrow$ 22.1%) and interaction projection (40.9% $\rightarrow$ 29.5%). The second gen-

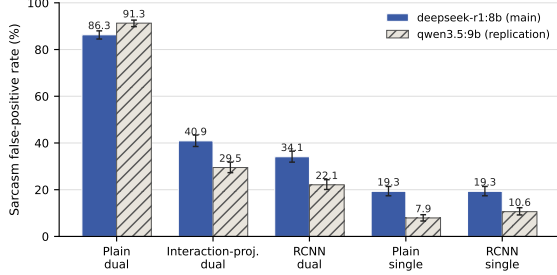


Figure 2: Sarcasm false-positive rates on the AI $\times$ non-sarcastic subset for five detectors and two generators (error bars are Wilson 95% confidence intervals; full numbers and paired tests in Appendix C.7).

erator thus reproduces the class-conditional failure; we have not tested broader generator families.

6 Detector–Judge Misalignment

To test whether detector scores track an independent generation-evaluation signal, we evaluate 600 base and SFT outputs from 300 prompts using the main-model detector and three blinded LLM judges (DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano; DeepSeek-AI, 2024; Gemini Team, 2025; OpenAI, 2026). The SFT model is Qwen3-8B (Qwen Team, 2025) fine-tuned on GenTrain with QLoRA (Detrmers et al., 2023); the protocol is detailed in Appendices B–C. Their rankings diverge: in Chinese, SFT raises the detector’s sarcasm score from 0.780 to 0.895 while reducing the mean judge score from 4.01 to 3.71. The detector also rates SFT outputs as more human-like in both languages, whereas the judges show the opposite direction. Pooled Spearman correlation is -0.212 for human-likeness and -0.020 for Chinese sarcasm.

The judges are LLMs, not human annotators, so what these results establish is detector–judge misalignment; how human readers would rate the same outputs is a question we do not answer. Even so, detector scores and an independent evaluation signal can rank the same outputs differently, and that alone argues against using detector scores as standalone generation rewards. The full protocol, per-judge tests, within-condition correlations, language analysis, and representative cases are reported in Appendices C–D.

7 Conclusion

Training sarcasm and AI-authorship detection together can fail in a way that overall metrics hide.

The plain dual-head model, matched in every other respect to its single-head counterpart, sharply overpredicts sarcasm on AI-generated non-sarcastic text; the failure persists with a second generator, and two intermediate modules reduce it. Representation analyses give the failure a name and a location—a source-correlated, style-entangled shortcut in a dominant-variance subspace—with causal attribution still open. On the generation side, SFT produced a language asymmetry, and detector and judge scores ranked the same outputs differently, so neither authorship nor sarcasm detector scores should be treated as standalone generation-quality signals. Whenever related judgments are learned through one shared textual representation, evaluation should be stratified by subgroup.

Limitations

DualTest combines human-authored texts with original sarcasm annotations and AI-generated texts with instruction-conditioned style labels. The AI-side sarcasm labels were not independently annotated by humans, so absolute scores on that side measure performance relative to the requested generation style. Replication with a second generator shows that the observed failure is not limited to deepseek-r1:8b, but does not establish generalization across broader open- and closed-source generator families. The source datasets also differ in their operational definitions of sarcasm, and the model does not use conversational context.

The mechanism evidence is primarily behavioral and linear. Source information is decodable and correlated with sarcasm decisions, yet the orthogonal high-variance control blocks any causal attribution to source identity. The buffering account of the intermediate modules is, for now, only descriptive: we have neither capacity-matched null modules nor a direct gradient analysis.

Finally, the generation analysis relies on three blinded LLM judges rather than human evaluators, with a 60:40 Chinese–English sample ratio; it supports a directional disagreement between detector and judge scores, not a claim about human-perceived generation quality or its precise language-specific magnitude. We also stopped short of reward-based generation. Folding the observed misalignment into reward design and measuring its dynamic effects under reinforcement learning is future work; so are causal-mediation and nonlinear tests for source attribution, and

parameter-isolating or data-side mitigations of the failure.

Ethical considerations

All human-authored texts used in this study come from publicly released research datasets (iSarcasm, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus) and are used in accordance with their original licenses. We collect no new user data and process no personally identifiable information. All AI texts were generated locally by open-source models (deepseek-r1.8b and qwen3.5.9b); they appear only in training and evaluation and are always labeled as machine-generated. The blinded LLM evaluation (DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano) consists of automated model-API calls and involves no human participants. The detectors are developed for research diagnosis. We explicitly warn (§6) that their scores should not be used directly as generation-quality or reward signals, nor should the detectors be deployed in high-risk settings such as content moderation without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here (the systematic misclassification of AI-generated non-sarcastic text) is itself an example of the systematic bias such systems can direct at particular content subgroups. AI-assisted tools were used during writing for language polishing and formatting; the study design, experiments, and conclusions are the authors’ own, and the authors take full responsibility for the work.

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A Detailed Dataset Statistics and Quality Checks

A.1 Source details. The standardized merged source collection contains 44,862 instances in total (86.9% English / 13.1% Chinese). Chinese–English 1:1 balance is achieved through bucketed sampling in the pipeline. The two English buckets and the Chinese non-sarcastic bucket contain more than 2,500 available instances and are randomly downsampled within each bucket. The Chinese sarcastic bucket contains 2,394 available unique instances and is oversampled within the bucket to

Source	Lang.	Inst.	Use
iSarcasm (SemEval-2022 T6)	English	3,467	Sarc. train + SFT
CSC	English	6,878	Sarc. train
ToSarcasm	zh 3,892 / en 6	3,898	Sarc. train + SFT
News Headlines v2	English	28,619	Sarc. train
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	Sarc. train

Table 4: Source datasets. Genre / annotation: iSarcasm — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

reach 2,500, producing the 4.24% duplication rate for that bucket reported in A.3.

Lang.	Authorship	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Three-dimensional distribution of DualTest.

A.2 Three-dimensional distribution of Dual-Test.

A.3 Data quality checks. All automated checks pass: language balance, label balance, balance across five folds, 99.97% completeness of human–AI pairing (9,996 / 9,999 pairs), and chain-of-thought (Wei et al., 2022) leakage detection. Train–test overlap comprises two SarcTrain instances (0.02%) and five AuthTrain instances (0.025%), all on the AI side of the test set; overlap on the human subset is zero. SarcTrain’s text-duplication rate is 0.34% for English / 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic bucket (4.24%; non-sarcastic bucket: 0.08%) and results from the oversampling used to fill that bucket (A.1).

A.4 AI-text generation protocol. All AI texts are produced by locally deployed generation models through the Ollama API (deepseek-r1:8b for the main results; qwen3.5:9b for cross-generator replication), using temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a generation limit of 500 tokens. Generation instructions use four

templates defined by language $\times$ style, with style determined by the seed instance’s sarcasm label. The templates explicitly constrain output style and length (20–80 Chinese characters; 10–50 English words); the length constraint also mitigates the confound of distinguishing human and AI texts by length (see the length-controlled analysis in Appendix E). The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

【任务】生成一条讽刺风格的评论
 【主题】{seed}
 【要求】
 - 直接给出简短的评论，不要解释思路或分析
 - 风格要讽刺、幽默、夸张
 - 表达与主题相关的观点即可，不必追求严谨
 - 20–80 字为佳
 【评论】

Its pinyin transliteration is:

[rènwù] shēngchéng yī tiáo fěngcì fēnggé de
 pínglùn
 [zhǔtí] {seed}
 [yāoqiú]
 - zhíjiē gěichū jiǎnduǎn de pínglùn, bùyào jiěshì
 sīlù huò fēnxī
 - fēnggé yào fěngcì, yōumò, kuāzhāng
 - biǎodá yǔ zhǔtí xiāngguān de guāndiǎn jíkě, bùbì
 zhuīqiú yánjǐn
 - 20–80 zì wéi jiā
 [pínglùn]

Its English translation is:

[Task] Generate a sarcastic-style
 comment
 [Topic] {seed}
 [Requirements]
 - Directly provide a short comment;
 do not explain your reasoning or
 analysis
 - The style should be sarcastic,
 humorous, and exaggerated
 - It is sufficient to express a
 viewpoint related to the topic;
 strict rigor is not required
 - Preferably 20–80 Chinese
 characters
 [Comment]

The Chinese non-sarcastic version changes only the style line to “The style should be normal and natural.” The two English templates parallel the Chinese templates (the style lines are “sarcastic, humorous, exaggerated” and “normal, natural,” respectively; length constraint: 10–50 words). Post-processing comprises three steps: removing `<think>...</think>` chain-of-thought segments; removing reasoning-preface terms such

as “好的” (*hǎo de* ‘okay’), “嗯” (*ń* ‘hmm’), “Okay,” “Let me,” and “First”; and removing Markdown markup. Failed generations are discarded. AuthTrain contains three generation failures (9,996 complete pairs; the corresponding human instances are retained). The AI side of the main DualTest set likewise excludes a small number of failures; cross-generator replication has zero failures (hence 1,550 AI $\times$ non-sarcastic instances for the replication set versus 1,513 for the main set). For cross-generator replication, thinking mode is explicitly disabled for the reasoning model (Ollama think = false) to prevent the reasoning process from exhausting the output budget. Inline chain-of-thought from deepseek-r1:8b is removed through the post-processing described above.

A.5 SFT training and generation-evaluation set.

GenTrain (2,428; 1,214 each in Chinese and English) is constructed by sampling approximately 40% of positive-sarcasm instances suitable for generation tasks and oversampling the minority language to achieve balance. By prompt type, it comprises 374 rewriting, 839 response, and 1,215 comment instances. Generation evaluation uses an independent set of 300 prompts (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. The detector scoring and three blinded LLM evaluations in §6 are both conducted on this set.

B Hyperparameters and Training Details

Group	Enc.	Inter.	Sarc.	Auth.
Plain / backbone	5e-7	—	5e-6	1e-5
Intermediate variants	5e-7	5e-6	5e-6	5e-6
Single-head	5e-7	5e-6 [†]	5e-6	1e-5

Table 6: Discriminative learning rates. [†]RCNN single-head only.

B.1 Discriminative learning rates. Note: the authorship-head learning rate is not identical across all settings (1e-5 for the plain dual-head and single-head models; 5e-6 for the architectural variants). This should be considered when interpreting cross-setting differences involving the authorship head. The sarcasm-head learning rate is 5e-6 in all settings, so conclusions on the sarcasm side are unaffected by this difference.

B.2 Shared training hyperparameters. Maximum 10 epochs; 200 warmup steps; weight decay 0.01; early-stopping patience 5 (normalized joint validation loss: $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{authorship validation loss} / \text{first-epoch value}]$); gradient clipping 1.0; physical batch size $4 \times$ gradient accumulation 4 = effective batch size 16; sampling ratio authorship:sarcasm = 2:1; sarcasm class weights [1.0, 2.5] (both the class weights and sampling ratio were specified a priori and not tuned on the test set); seed 42 (reinitialization with seed + fold index for each fold; all architecture/ablation experiments share the fold assignments and seed scheme); fixed 90/10 authorship-data holdout shared across all folds. Final model: retrained on all of SarcTrain and all of AuthTrain for a number of epochs equal to the mode of the best early-stopping epochs across the five runs +1 (maximum 10), with an independent seed (seed +1000), no validation set, and no early stopping. All main tables report the five-run mean $\pm$ standard deviation rather than a point estimate from the final model.

Software and hardware environment. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB); PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4); blinded LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.3 SFT (QLoRA). Qwen3-8B; 4-bit loading (bfloat16 computation); LoRA (Hu et al., 2022a) rank 64 / alpha 16 / dropout 0; target modules q/k/v/o_proj and gate/up/down_proj; maximum sequence length 2,048; gradient checkpointing enabled; learning rate $5e-5$; 3 epochs; batch size $1 \times$ gradient accumulation 8; warmup ratio 0.05; best checkpoint selected by validation loss.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). Main-model macro-averaged F1, under the same five-run protocol and calculated from each run’s confusion matrix: sarcasm **0.6678 $\pm$ 0.036** and authorship **0.9298 $\pm$ 0.005**. Macro-averaged F1 is higher than positive-class F1 (0.6309) because the majority class has a higher F1; the ordering in §5.1 remains unchanged under both conventions.

C.2 Complete four-quadrant results for the authorship head (including single-head counterfactuals). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual	91.4	91.5	3.6	6.1
Inter.-proj. dual	85.4	85.4	1.8	4.1
RCNN dual (main)	94.8	93.2	5.5	6.6
Plain single	90.4	86.6	4.1	4.2
RCNN single	90.5	92.4	3.6	5.7

Table 7: Authorship-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

The single-head authorship counterfactuals perform at the same level as the dual-head models (differences within single-digit pp in every quadrant). Joint training induces no class-conditional failure on the authorship side, consistent with the directionality claim in §5.3. The authorship head is likewise stable on the cross-generator replication set (qwen3.5:9b): AI $\times$ non-sarcastic false-positive rates are 0.9% / 1.3% / 0.5% for plain / interaction-projection / RCNN, respectively; AI $\times$ sarcastic false-positive rates are all $\leq 0.4\%$; and overall authorship F1 scores are 0.952 / 0.917 / 0.966, respectively.

C.3 Blinded-evaluation protocol, inter-judge agreement, and group-level tests (§6). Protocol: each sample is sent to each of the three judge APIs independently with a single prompt (May 2026; temperature = 0.3; GPT-5-nano does not support a temperature parameter and uses its default decoding). The prompt contains only the input and the text to be evaluated, without ground-truth source information, and asks the judge to infer the source and then provide two scores from 1 to 5 for sarcasm and human-likeness. Base and SFT outputs are evaluated independently with the same prompt. Pairwise instance-level agreement between judges is moderate to low: Spearman ρ is 0.20–0.38 for sarcasm (all $p < 10^{-6}$) and 0.03–0.23 for human-likeness (one pair not significant). The misalignment conclusion in §6 therefore does not rest on instance-level rank agreement, but on consistent condition-level directions. As shown below, “SFT sarcasm is below the base model” is independently significant in five of six groups across

2 languages $\times$ 3 judges under Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995). The pooled correlation analysis (Table 8) uses the mean of the three judges, which can partly average out inter-judge noise.

Subset	n	Sarc. ρ	Auth. ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 8: Spearman correlations between detector and blinded LLM judgments (main-model detector, $n = 600$ pooled over base and SFT). * significant; other entries not significant.

Judge / lang.	n	SFT	Base	adj. p
DeepSeek-V3 / en	120	3.292	4.025	<.0001
DeepSeek-V3 / zh	180	3.878	4.044	.0008
Gemini-2.5-FL / en	120	3.750	3.992	.0008
Gemini-2.5-FL / zh	180	3.994	4.028	.2008*
GPT-5-nano / en	119	3.681	4.084	<.0001
GPT-5-nano / zh	180	3.517	3.911	<.0001

Table 9: Group-level tests for blinded sarcasm judgments (Wilcoxon + BH-FDR). *not significant.

C.4 Paired tests between SFT and human references (supplementary). This test is performed on the Chinese subset of the generation-evaluation set ($n = 180$; GPT-5-nano is missing three instances, leaving 177). Directions are inconsistent: for DeepSeek-V3, SFT is significantly below the reference on human-likeness (-0.161 , $p = 0.0025$); for GPT-5-nano, SFT is significantly above the reference on sarcasm ($+0.186$, $p = 0.0114$); the other four groups are not significant.

C.5 Plain-detector control (robustness).

Scores: Chinese sarcasm $0.765 \rightarrow 0.763$ (-0.001 ; no change), English sarcasm $0.843 \rightarrow 0.659$ (-0.184); authorship increases by $+0.760$ in Chinese and $+0.705$ in English. The two detectors agree closely on the rise in authorship scores, while the interpretation of the sarcasm direction depends on the detector. Correlations: overall sarcasm $\rho = +0.142$ / authorship -0.214 ; Chinese sarcasm -0.037 (not significant) / authorship -0.209 ; English sarcasm $+0.342$ / authorship -0.283 . Differences from the main

model are ≤ 0.02 for every subset, showing that the misalignment is robust to detector architecture.

Within-condition correlations (main-model detector, recomputed separately within the base and SFT conditions). Authorship dimension: within the base condition, $\rho = +0.03$ ($n = 300$, not significant; Chinese $+0.10$ / English -0.09 , neither significant); within the SFT condition, -0.12 ($p = 0.039$; Chinese -0.19 , $p = 0.009$ / English -0.12 , not significant). The pooled-sample correlation of -0.212 is driven mainly by the opposite between-condition changes (SFT increases the detector score by approximately 0.67 while decreasing the mean blinded judgment); there is no positive alignment within conditions. Sarcasm dimension: within-condition correlations for Chinese are base $+0.05$ (not significant) / SFT $+0.15$ ($p = 0.048$), and for English are base $+0.41$ / SFT $+0.37$ (both significant). The Chinese within-condition correlation is at most weakly positive, consistent with the near-zero pooled result.

C.6 Detector scores before and after SFT and language asymmetry. Scores from the main-model detector for base and SFT outputs (0–1):

Lang.	Cond.	Sarc.	Auth.
Chinese	Base $\rightarrow$ SFT	0.780 $\rightarrow$ 0.895	0.289 $\rightarrow$ 0.969
English	Base $\rightarrow$ SFT	0.854 $\rightarrow$ 0.730	0.198 $\rightarrow$ 0.864

Table 10: Detector scores before/after SFT (main-model detector).

Sarcasm scores show language asymmetry (Chinese $+0.115$ / English -0.124); the 60:40 language imbalance in the SFT training samples may partly explain this directional difference (see Limitations). Mean judgments from the three LLMs (sarcasm / human-likeness) are 3.77/4.17 for the human reference, 4.01/4.24 for the base model, and 3.71/4.08 for SFT. Authorship scores rise sharply in both languages, meaning that the detector judges SFT outputs as “more human-like”; the conflict between this judgment and the direction of blinded judgments is analyzed in §6.

C.7 Quantifying test-set uncertainty: confidence intervals, paired tests, and subset F1.

This section quantifies sampling uncertainty over test instances (Wilson 95% intervals for quadrant rates; 95% percentile intervals from instance-level bootstrap with 10,000 resamples for F1), complementing the between-run standard deviations re-

ported in Table 2 and Table 1. Models are the final models for each architecture, and the reporting convention follows Table 3. The five detectors are trained independently; their quadrant false-positive rates have a consistent ordering that replicates with the second generator. Between-run variation in overall F1 (Table 1, ± 0.02 – 0.03) is also far smaller than the differences of tens of percentage points across quadrants.

Model	deepseek	qwen
Plain dual	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Inter.-proj. dual	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual (main)	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI $\times$ non-sarcastic subset (two generators; %).

Intervals do not overlap between any dual-head and single-head model, nor between the plain dual-head model and either intermediate-layer dual-head model. Exact paired McNemar tests on the same texts (main-generator set, AI $\times$ non-sarcastic subset) yield: plain dual-head vs. plain single-head, discordant pairs 1,021 / 7, $p < 10^{-291}$; RCNN dual-head vs. RCNN single-head, 239 / 15, $p < 10^{-52}$; plain dual-head vs. RCNN dual-head, 805 / 15, $p < 10^{-215}$; interaction-projection dual-head vs. RCNN dual-head, 203 / 100, $p = 3.3 \times 10^{-9}$. All four differences are significant.

Main-model authorship F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the authorship label is constant within the human subset, positive-class F1 degenerates to the recall convention and is not separately reported. Chinese sarcasm F1 is substantially higher than English sarcasm F1 (0.819 vs. 0.429 for the main model) for two direct reasons. First, sarcasm prevalence is 63.8% in the Chinese subset but only 14.2% in the English subset. Positive-class F1 increases with prevalence, so the values for the two languages are not directly comparable. Second, the source benchmarks differ in difficulty: the English human side is dominated by intent-oriented iSarcasmEval tweets (the best systems on that benchmark achieve F1 of approximately 0.56–0.61; §2.1), whereas the Chinese side uses topic-comment-style ToSarcasm. Regarding the encoder, DeBERTa-v3’s SentencePiece (Kudo and Richardson, 2018) vocabulary tokenizes Chinese

largely at the character level (measured at approximately 1.02 tokens per character, with approximately 2.5% out-of-vocabulary symbols). The Chinese subset nevertheless achieves high F1 under these conditions, consistent with the dominant roles of prevalence and genre. The difference between overall and human-subset F1 (0.663 vs. 0.629 for the main model) comes from the contribution of AI-side instances; accordingly, the metric clarification in §5.1 notes that the overall value aggregates both source sides. On the human subset, dual- versus single-head F1 differences are within 0.03 and vary in direction. The visible loss from negative transfer on the sarcasm side is concentrated in the AI-side quadrants, consistent with the class-conditional characterization in §5.3.

False-positive rates use a threshold of 0.5. Instance-level score distributions show that the failure is not a threshold artifact: for the plain dual-head model on the AI $\times$ non-sarcastic subset, the median sarcasm probability is 0.767 (interquartile range 0.635–0.839), so the entire distribution shifts rightward across the threshold. The single-head medians are 0.204 (plain) / 0.069 (RCNN). At the same time, AUC values for ranking sarcasm probability within the AI side are similar across models (0.82–0.88): joint training mainly raises the level of sarcasm scores on AI texts while largely preserving their internal ordering, consistent with the characterization that a source-related component is added to the sarcasm signal.

D Representative Cases of Detector–LLM Score Misalignment

Cases are selected from the paired table of main-model detector scores and blinded judgments from the three LLMs by the magnitude of disagreement (detector scores: 0–1; LLM scores: mean of three evaluators on a 1–5 scale; long texts truncated). Chinese examples retain the original model input and provide a pinyin transliteration and an English translation in brackets.

D.1 Classified as “human-like” by the detector but “not human-like” by the LLMs (authorship misalignment; corresponding to the mechanism discussed in §6). In the first case, machine-generated text with obvious chain-of-thought traces is nevertheless classified as human by the authorship head with confidence 1.00 (see §6 for the mechanism interpretation).

Model	Full set	Chinese	English	Human subset	Human-zh	Human-en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head (main)	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 12: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for the main model).

L	S	$P_{\text{hum}}/\text{LLM Text (excerpt)}$
zh	SFT	1.00 / 2.67 “好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [hǎo, wǒ xūyào zhēnduì Sānxīng huíduì Huáwéi de xīnwén shēngchéng yī tiáo fěngcǐ pínglùn. shǒuxiān...] [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (chain-of-thought leakage)
zh	SFT	1.00 / 2.33 “跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈” [pǎofēn jìngsài: cānshù jiāolǜ: pǎofēn jiāolǜ: shuǐhuò: hāhāhā] [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
en	SFT	1.00 / 2.67 “Same as me, I’m not sure” (vacuous short sentence)

Table 13: Authorship misalignment cases. $P_{\text{hum}} = P(\text{human})$.

L	S	$P_{\text{sar}}/\text{LLM Text (excerpt)}$
zh	SFT	1.00 / 2.67 “三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sānxīng: wǒ cái shì dì-yī! nǐ kànkàn wǒ zhè shèxiàngtóu zǒngfēn, 109+96, 112+89, wǒ yíng le!] [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
zh	SFT	0.99 / 2.67 “恭喜恭喜，美国牛牛牛，继续加油” [gōngxǐ gōngxǐ, Měiguó niú niú niú, jìxù jiāyóu] [“Congratulations, congratulations, America is awesome awesome awesome, keep it up!”] (formulaic mock-praise)
en	SFT	0.93 / 2.67 “2 hours left to tip it over, I’m going to the toilet”

Table 14: Sarcasm false-positive cases. $P_{\text{sar}} = P(\text{sarcastic})$.

D.2 Classified as “sarcastic” by the detector but “not sarcastic” by the LLMs.

D.3 Classified as “sarcastic” by the LLMs but missed by the detector. Taken together, D.2 and D.3 suggest that the sarcasm head may favor surface sarcasm markers in the training distribution (mock-praise formulas, exclamation marks, and colloquial short sentences) while being insensitive to rhetorically complete written irony that depends on semantic reversal; the LLM judges show the opposite tendency. Because the cases are selected by disagreement magnitude, this observation is a qual-

L	S	$P_{\text{sar}}/\text{LLM Text (excerpt)}$
zh	Base	0.14 / 4.00 “特朗普承诺 120 亿补贴撒大饼，农民收到的却是缩水成硬币的承诺……” [Tèlǎngpǔ chéngnuò 120 yì bǔtiē sā dà bǐng, nóngmín shōu dào de què shì suōshuǐ chéng yìngbì de chéngnuò...] [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
zh	Base	0.24 / 4.00 “特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [Tèsilā: jiàgé tiàoshuǐ hòu yòu tiàoshuǐ, xiāofèizhě: qiánbāo tiàoshuǐ hòu yòu tiàoshuǐ.] [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
en	Base	0.12 / 4.00 “Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis + ironic emoji)

Table 15: Sarcasm false-negative cases. $P_{\text{sar}} = P(\text{sarcastic})$.

itative annotation of the near-zero Chinese sarcasm correlation in §6 rather than a frequency estimate. Systematic testing is left to future work.

E Preliminary Representation-Level Analysis of the Failure Mechanism

This appendix records the complete protocol, data, and self-checks for the representation-level tests in §5.3.2. The interpretation of conclusions follows §5.3.2.

E.1 Setup. All analyses use the frozen model that produces the 86.3% result for the plain dual-head model in the main text, without updating its parameters. Direction estimation and metric evaluation are cross-fitted (the direction is estimated on the fitting half and metrics are calculated on the evaluation half), so the baseline false-positive rate in this appendix (84.8% on the evaluation half) differs slightly from the full-set value in the main text (86.3%). Code self-checks: manually computed logits match model outputs elementwise (maximum error 1.4×10^{-6}); the erasure-projection residual is approximately 10^{-6} ; LEACE satisfies its defining property (source AUC on the fitting half: $0.988 \rightarrow 0.042$); and probe selectivity passes

(true labels: 0.986; permuted labels: 0.512).

E.2 Protocol. We record here the principles, citations, and implementation details for each test underlying §5.3.2. The linear probe is logistic regression with five-fold cross-validation. The selectivity control retraines the same probe with permuted source labels. Correlation analyses are rechecked after stratification by text-length tertile. Alignment is the directional cosine under whitened geometry ($\Sigma^{1/2}w$). Erasure interventions include a single direction, the INLP subspace ($k = 1-16$; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Placebos comprise ten random whitened directions and erasure after re-estimating directions from permuted labels. srcORTH-PC1 is the maximum-variance direction after orthogonalization against the source direction; erasing it does not alter source decodability and is used to distinguish the effects of “removing the source concept” and “perturbing the principal-variance structure.”

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC 0.986 (sarcastic subset: 0.990 / non-sarcastic subset: 0.985; permuted-label control: 0.512; single-head model: 0.974, meaning that decodability is not unique to joint training). On AI $\times$ non-sarcastic instances, the correlation between source logit and sarcasm logit is $\rho = 0.519$ (0.44/0.56/0.63 after controlling by text-length tertile). The sarcasm decision direction and source direction are nearly orthogonal in the original geometry ($\cos \approx -0.03$) but moderately aligned in whitened geometry ($|\cos| = 0.41$ when measured using the probe direction / 0.49 using the authorship-head direction). All are compatible with the shortcut hypothesis, but all constitute correlational evidence.

E.4 Intervention evidence. gap = AI $\times$ non-sarcastic false-positive rate – human $\times$ non-sarcastic false-positive rate; source AUC is the decoding ability of a probe retrained after erasure.

Interpretation: single-direction and low-dimensional INLP erasure are ineffective. LEACE initially appears effective (source is indeed erased, the gap falls from 52 to 19, and the placebos are all ineffective). However, erasing a high-variance direction orthogonal to source (leaving source information entirely intact) likewise reduces the false-positive rate to 66.7 and the gap to 12.6, an effect almost identical to that of LEACE.

Erasure scheme	AI $\times$ n FP	H $\times$ n FP	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈ 0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (full)	68.7	49.5	19.1	0.589
Random	84.9	32.7	—	0.982
whitened $\times 10$				
srcORTH-PC1	66.7	54.1	12.6	0.982

Table 16: Intervention evidence. H $\times$ n = human $\times$ non-sarcastic; AI $\times$ n = AI $\times$ non-sarcastic; FP in %.

E.5 Supplementary observation. The mean false-positive rate across the two non-sarcastic source sides is almost unchanged before and after the intervention (58.8 $\rightarrow$ 59.1). The intervention does not “fix” the model; it merely transfers part of the overprediction of sarcasm on the AI side to the human side.